The empirical recurrence for OEIS A224744: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A224744 records "Empirical recurrence of order 55", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 312$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 55, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 55 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A224744 is "Number of (n+1)X8 0..1 matrices with each 2X2 permanent equal.". It has offset 1 and begins

$$3410, 170754, 4028161, 149592266, 4390392346, 146736389761, 4577116826010, 148230752711146, \ldots$$

The entry states:

> Empirical recurrence of order 55 (see link above)

As of the "Last modified" line on the live entry (Oct 03 2025 23:36:25, revision 9) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 312$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 312.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 624$ exact terms of the model returns order 55 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{55} c_i\, a(n-i),$$

c_1	48	c_{15}	3964546897451937	c_{29}	−24509776645981526751694	c_{43}	−46729449
c_2	−24	c_{16}	8681357395137836	c_{30}	139570162331843666394522	c_{44}	98455693
c_3	−25429	c_{17}	−100151521896214590	c_{31}	29659261289570134585868	c_{45}	23274651
c_4	249600	c_{18}	−59193573623950180	c_{32}	−510102061006196734465896	c_{46}	−9688998
c_5	4136014	c_{19}	1794915980022787224	c_{33}	137659045977638251002786	c_{47}	−6716071
c_6	−66545398	c_{20}	−793452107572240912	c_{34}	1353282358848681990405012	c_{48}	41755158
c_7	−229014820	c_{21}	−23421050984594666996	c_{35}	−898675393739804884831572	c_{49}	92035094
c_8	7754341688	c_{22}	28436886246974409000	c_{36}	−2538628075423822788633573	c_{50}	−9202407
c_9	−6664841518	c_{23}	224462089791291975336	c_{37}	2653060422529980580645044	c_{51}	−984649
c_{10}	−508973605180	c_{24}	−424033900038336001719	c_{38}	3185083149320739317845680	c_{52}	9679730
c_{11}	1628611719916	c_{25}	−1568901616914705349704	c_{39}	−4880579238043480446738345	c_{53}	−915993
c_{12}	20756456235325	c_{26}	4093413980735911127464	c_{40}	−2309628550387630425595524	c_{54}	−355433
c_{13}	−105282909067100	c_{27}	7748235686573801992573	c_{41}	5903861666917943927998206	c_{55}	485996
c_{14}	−542511759403584	c_{28}	−27992808801408486001384	c_{42}	354255725915473038914592		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 55, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $55+4$ terms removed returns order 55 as well, so $\deg q_{\min} = 55 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $55+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 55 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A224744.

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